



L1B: Runology

CS1101S: Programming Methodology

Boyd Anderson

14th August, 2026

Version 1.0 — Built: 2026-08-19 21:18:42



First Mission

Recap

Python in Pictures



Announcements



Announcements

- ▶ This Monday/Tuesday 2-h **Studio** session S2 on Elements of Programming



Announcements

- ▶ This Monday/Tuesday 2-h **Studio** session S2 on Elements of Programming
- ▶ Don't worry if you have not been assigned a session. We will have an online option on Tuesday at 6pm-8pm.



Announcements

- ▶ This Monday/Tuesday 2-h **Studio** session S2 on Elements of Programming
- ▶ Don't worry if you have not been assigned a session. We will have an online option on Tuesday at 6pm-8pm.
- ▶ Don't forget about our Ed forums:
<https://edstem.org/us/courses/101145/discussion>



First Mission

Recap

Python in Pictures



First Mission: Rune Trials



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.
- ▶ Missions are submitted individually and carry significant XP.



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.
- ▶ Missions are submitted individually and carry significant XP.
- ▶ Path P1B will be released at 11:45 today.



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.
- ▶ Missions are submitted individually and carry significant XP.
- ▶ Path P1B will be released at 11:45 today.
- ▶ You *can* discuss paths, missions, and quests with your friends and avengers.



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.
- ▶ Missions are submitted individually and carry significant XP.
- ▶ Path P1B will be released at 11:45 today.
- ▶ You *can* discuss paths, missions, and quests with your friends and avengers.
- ▶ You *must* do the actual programming yourself. Do *not* copy anyone's programs, from other students or from previous batches or using AI (without attribution).



First Mission: Rune Trials

- ▶ ...will be released at 11:45 **today**.
- ▶ Missions are submitted individually and carry significant XP.
- ▶ Path P1B will be released at 11:45 today.
- ▶ You *can* discuss paths, missions, and quests with your friends and avengers.
- ▶ You *must* do the actual programming yourself. Do *not* copy anyone's programs, from other students or from previous batches or using AI (without attribution).
- ▶ Submitting other people's work as your own is a serious offence and will be severely punished by NUS.



First Mission

Recap

Evaluating combinations, 1.1.3

Python in Pictures



Recap: Elements of Programming (1.1)



Recap: Elements of Programming (1.1)

- ▶ Primitives: things like `-42` or `False`



Recap: Elements of Programming (1.1)

- ▶ Primitives: things like `-42` or `False`
- ▶ Combination: things like `-42 * 7`



Recap: Elements of Programming (1.1)

- ▶ Primitives: things like `-42` or `False`
- ▶ Combination: things like `-42 * 7`
- ▶ Name abstraction: things like `size = 2`



Recap: Elements of Programming (1.1)

- ▶ Primitives: things like `-42` or `False`
- ▶ Combination: things like `-42 * 7`
- ▶ Name abstraction: things like `size = 2`
- ▶ Conditional expressions:
consequent `if` predicate `else` alternative



Recap: Elements of Programming (1.1)

- ▶ Primitives: things like `-42` or `False`
- ▶ Combination: things like `-42 * 7`
- ▶ Name abstraction: things like `size = 2`
- ▶ Conditional expressions:
consequent `if` predicate `else` alternative
- ▶ Functional abstraction: more on this today



A function can be a “black box”

Example

```
math_sqrt
```



A function can be a “black box”

Example

`math_sqrt`

- ▶ Input: any number
- ▶ Output: a number whose square is the input number



A function can be a “black box”

Example

`math_sqrt`

- ▶ Input: any number
- ▶ Output: a number whose square is the input number

Function application

```
print(math_sqrt(15))
```





Means of Abstraction: Compound functions

Example

```
def square(x):  
    return x * x
```





Means of Abstraction: Compound functions

Example

```
def square(x):  
    return x * x
```



What does this statement mean?

Like declaration assignments, this *function definition* declares a name, here `square`. The value associated with the name is a function that takes an argument `x` and produces (returns) the result of multiplying `x` by itself.



Evaluation of expressions

1 + 2 * 3 + 4





Evaluation of expressions

1 + 2 * 3 + 4



Replace “eligible” subexpression with result

(1 + (2 * 3)) + 4



Evaluation of expressions

1 + 2 * 3 + 4



Replace “eligible” subexpression with result

(1 + (2 * 3)) + 4

(1 + 6) + 4



Evaluation of expressions

$1 + 2 * 3 + 4$



Replace “eligible” subexpression with result

$(1 + (2 * 3)) + 4$

$(1 + 6) + 4$

$7 + 4$



Evaluation of expressions

`1 + 2 * 3 + 4`



Replace “eligible” subexpression with result

`(1 + (2 * 3)) + 4`

`(1 + 6) + 4`

`7 + 4`

`11`



Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```





Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```



Evaluate what is after the =

```
size = 3  
2 * size
```



Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```



Evaluate what is after the =

```
size = 3  
2 * size
```

Then replace that name in rest of the program with that value.

```
2 * 3
```



Evaluation of declaration assignments

```
a = 3 * 5  
print(a + 5)
```



Evaluate what is after the =

```
size = 3  
2 * size
```

Then replace that name in rest of the program with that value.

```
2 * 3
```

```
6
```



Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```





Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```



Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```

If the predicate evaluates to True, evaluate the consequent

```
1
```



Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```

If the predicate evaluates to `True`, evaluate the consequent

1

If the predicate evaluates to `False`, evaluate the alternative



Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```

If the predicate evaluates to True, evaluate the consequent

```
1
```

If the predicate evaluates to False, evaluate the alternative

```
1 if 3 > 4 else 2
```





Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```

If the predicate evaluates to True, evaluate the consequent

```
1
```

If the predicate evaluates to False, evaluate the alternative

```
1 if 3 > 4 else 2
```

```
1 if False else 2
```





Evaluation of conditional expressions

```
1 if 3 < 4 else 2
```



Evaluate the predicate first.

```
1 if True else 2
```

If the predicate evaluates to True, evaluate the consequent

```
1
```

If the predicate evaluates to False, evaluate the alternative

```
1 if 3 > 4 else 2
```



```
1 if False else 2
```

```
2
```



First Mission

Recap

Python in Pictures



Primitives

Some **simple** **runes** such as

- ▶ `rcross`
- ▶ `sail`
- ▶ `corner`
- ▶ `nova`
- ▶ `heart`
- ▶ `show` is a primitive operation that displays a rune.

```
show(heart)
```





Where do these primitives come from?

The `rune` module

`rcross`, `sail`, `corner`, `nova`, `heart`, and `show` are not built into Python. They come from the `rune` module.



Where do these primitives come from?

The `rune` module

`rcross`, `sail`, `corner`, `nova`, `heart`, and `show` are not built into Python. They come from the `rune` module.

Import them before use

Every program that uses them needs an `import` statement first:

```
from rune import (  
    show, heart, rcross, sail, nova, corner  
)
```





Primitive Operations

Rune-Transformations

Functions that take a rune and produce a new rune from it

Example rune transformation

Turn a given rune a quarter turn right: `quarter_turn_right`

Example in Python

```
quarter_turn_right(quarter_turn_right(sail))
```





Abstraction: Functions and naming

```
def turn_upside_down(rune):  
    return quarter_turn_right(quarter_turn_right(rune))  
  
# example use:  
show(turn_upside_down(heart))
```





Abstraction: Functions and naming (continued)

```
def quarter_turn_left(rune):  
    return quarter_turn_right(turn_upside_down(rune))  
  
# example use:  
show(quarter_turn_left(heart))
```





Combination operation: **stacking**

```
stack(rcross, sail)
```





Your turn: Combination abstraction

```
def beside(rune1, rune2):  
    ...  
  
# example use:  
show(beside(rcross, sail))  
# should show rcross on the left  
#           and sail on the right
```



Solution: Combination abstraction

```
def beside(rune1, rune2):  
    return quarter_turn_left(  
        stack(  
            quarter_turn_right(rune1),  
            quarter_turn_right(rune2)  
        )  
    )
```





Solution: Combination abstraction

```
def beside(rune1, rune2):  
    return quarter_turn_left(  
        stack(  
            quarter_turn_right(rune1),  
            quarter_turn_right(rune2)  
        )  
    )
```



Good news

`beside` is important and it's already pre-defined for you in the `rune` module!



Time for the Check-In!



Time for the Check-In!

Let's DDOS the Source Academy!



Time for the Check-In!

Let's DDOS the Source Academy!

Look for Check-In I1B in the Source Academy!



Time for the Check-In!

Let's DDOS the Source Academy!

Look for Check-In I1B in the Source Academy!

(Only your best 18 of 20 Check-Ins count, so you may miss up to two without penalty.)



Combination: stacking something n times

```
stackn(5, heart)
```



Lecture L2A

We will define `stackn` in Python, using rune primitives.



Naming and more combination: rectangular quilting

```
my_quilt = stackn(  
    5,  
    quarter_turn_right(stackn(7,  
        quarter_turn_left(heart)))  
)
```



Abstraction: rectangular quilting

```
def quilt(n, m, rune):  
    return stackn(  
        n,  
        quarter_turn_right(stackn(m,  
            quarter_turn_left(rune)))  
    )
```





Combination: a more complex rune

```
stack(  
  beside(quarter_turn_right(rcross),  
        turn_upside_down(rcross)),  
  beside(rcross,  
        quarter_turn_left(rcross))  
)
```





Naming interesting things

```
my_cross = stack(  
    beside(quarter_turn_right(rcross),  
           turn_upside_down(rcross)),  
    beside(rcross,  
           quarter_turn_left(rcross))  
)
```





Abstraction: making a cross from any rune

```
def make_cross(rune):  
    return stack(  
        beside(quarter_turn_right(rune),  
                turn_upside_down(rune)),  
        beside(rune,  
                quarter_turn_left(rune))  
    )
```





Abstraction: making a cross from any rune

```
def make_cross(rune):  
    return stack(  
        beside(quarter_turn_right(rune),  
                turn_upside_down(rune)),  
        beside(rune,  
                quarter_turn_left(rune))  
    )
```

`make_cross` is a transformation.



Combination: repeating a transformation

```
show(make_cross(make_cross(nova)))
```





Abstraction: repeating a transformation n times

Pre-defined in module `repeat`:

```
show(repeat_apply(make_cross, 5, rcross))
```



Lecture L2A

Like `stackn`, we will define this abstraction in Python next Wednesday!



Recap

- ▶ We started with primitive constants.



Recap

- ▶ We started with primitive constants.
- ▶ No idea how they are rendered!



Recap

- ▶ We started with primitive constants.
- ▶ No idea how they are rendered!
Example: `heart`



Recap

- ▶ We started with primitive constants.
- ▶ No idea how they are rendered!
Example: `heart`
- ▶ We introduced primitive combinations.



Recap

- ▶ We started with primitive constants.
- ▶ No idea how they are rendered!
Example: `heart`
- ▶ We introduced primitive combinations.
- ▶ No idea how the primitive combinations work!



Recap

- ▶ We started with primitive constants.
Example: `heart`
- ▶ We introduced primitive combinations.
Example: `quarter_turn_right`



Recap

- ▶ We started with primitive constants.
Example: `heart`
- ▶ We introduced primitive combinations.
Example: `quarter_turn_right`
- ▶ Yet we can use primitives and combinations to generate complex runes.



Recap

- ▶ We started with primitive constants.
 - ▶ No idea how they are rendered!
Example: `heart`
- ▶ We introduced primitive combinations.
 - ▶ No idea how the primitive combinations work!
Example: `quarter_turn_right`
- ▶ Yet we can use primitives and combinations to generate complex runes.
 - **Abstractions** allow us to master complexity:



Recap

- ▶ We started with primitive constants.
 - ▶ No idea how they are rendered!
Example: `heart`
- ▶ We introduced primitive combinations.
 - ▶ No idea how the primitive combinations work!
Example: `quarter_turn_right`
- ▶ Yet we can use primitives and combinations to generate complex runes.
 - **Abstractions** allow us to master complexity:
Naming



Recap

- ▶ We started with primitive constants.
 - ▶ No idea how they are rendered!
Example: `heart`
- ▶ We introduced primitive combinations.
 - ▶ No idea how the primitive combinations work!
Example: `quarter_turn_right`
- ▶ Yet we can use primitives and combinations to generate complex runes.
 - **Abstractions** allow us to master complexity:
Naming and **functional abstraction**



Happy Half Valentine's Day!

```
from rune import show, heart,
    translate, rotate, red,
    pink, overlay, scale

a = translate(-0.18, 0.02,
    rotate(0.3,
    red(scale(0.42, heart))))
b = translate(0.18, -0.02,
    rotate(0.3,
    pink(scale(0.42, heart))))

show(overlay(a, b))
```

